Higher derivatives

1 Suppose $y = \dfrac{2}{x^5}$. Find:

a $\dfrac{dy}{dx}$ **b** $\dfrac{d^2y}{dx^2}$ **c** $\dfrac{d^3y}{dx^3}$

2 Suppose $y = 5x^8$. Find:

a $\dfrac{dy}{dx}$ **b** $\dfrac{d^2y}{dx^2}$ **c** $\dfrac{d^3y}{dx^3}$ **d** $\dfrac{d^4y}{dx^4}$

e $\dfrac{d^5y}{dx^5}$

3 Consider the function $y = -2x^4 + 4x^3 - 7x^2 + 8x - 9$. Find:

a $\dfrac{dy}{dx}$

b The value of $\dfrac{dy}{dx}$ when $x = 3$

c $\dfrac{d^2y}{dx^2}$

d The value of $\dfrac{d^2y}{dx^2}$ when $x = 4$

e $\dfrac{d^3y}{dx^3}$

f The value of $\dfrac{d^3y}{dx^3}$ when $x = 6$

4 Find the second derivative of the following functions:

a $y = x^4 - 6x^3 + 8x^2 - 10$

b $f(x) = x^{\frac{1}{7}}$

c $f(x) = (3x + 2)^4$

d $f(x) = \dfrac{4}{(x + 8)^2}$

e $y = (7 - 6x^2)^{\frac{1}{3}}$

f $y = 4\sqrt{6 - x}$

g $f(x) = x^2(x^3 + 3)$

h $f(x) = (x^4 + 2)(3 - 2x^3)$

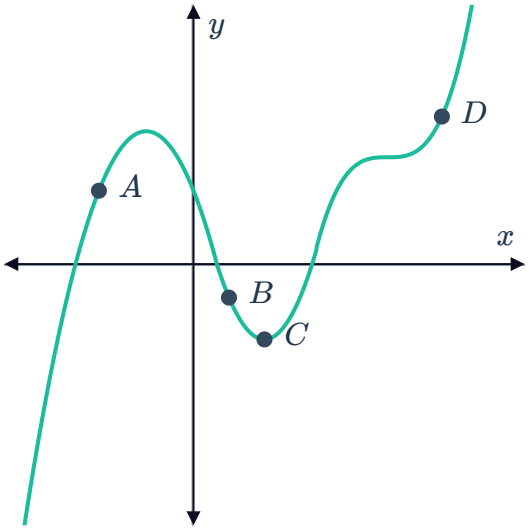
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Concavity

5 Consider the given function $y = f(x)$ with points A, B, C and D:

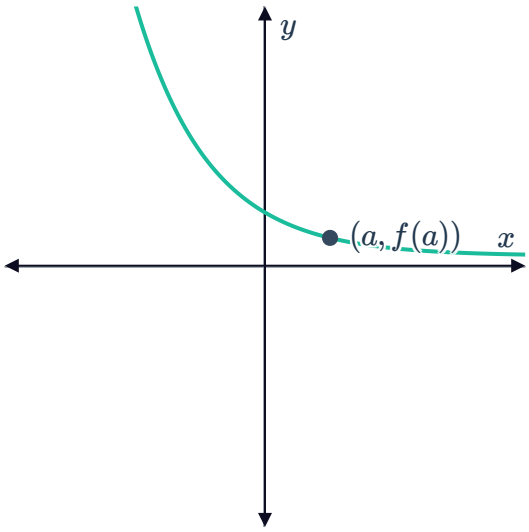
a At which of these points is $f''(x) > 0$?

b At which of these points is $f''(x) < 0$?

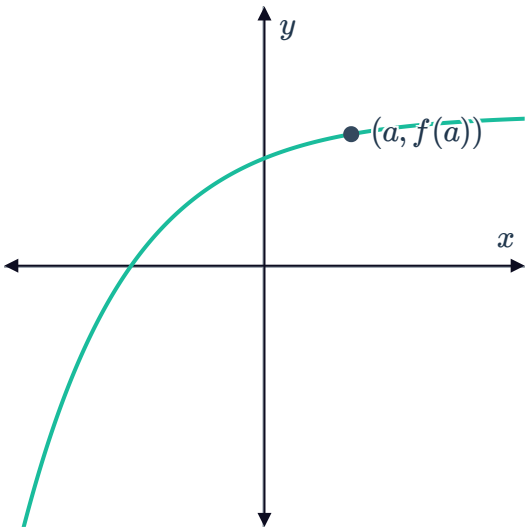


6 For each of the following curves, state the sign of $f'(a)$ and $f''(a)$:

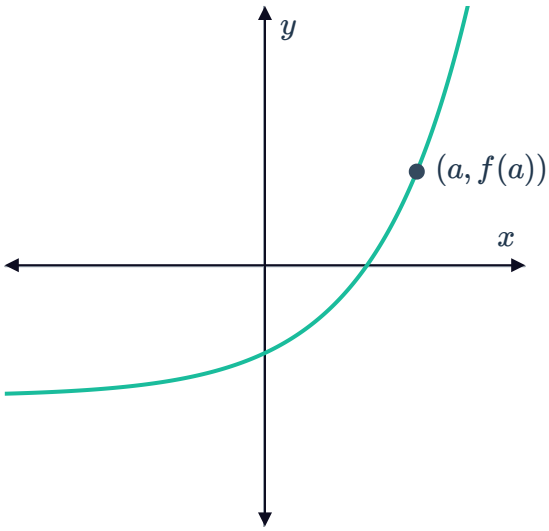
a



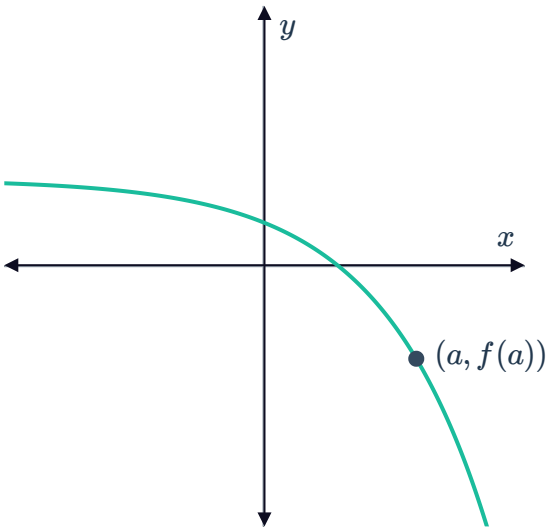
b



c



d



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7 For each of the following quadratic functions:

i State whether the leading coefficient of $f(x)$ is positive or negative.

ii Hence, determine the nature of the turning point.

iii Find $f'(x)$.

iv Find $f''(x)$.

v State whether the curve is concave up or down.

a $f(x) = x^2 - 4x + 9$

b $f(x) = -x^2 + 4x - 9$

8 Consider the function $f(x) = 4x^2 + 3x + 2$.

a Find the first derivative.

b Find the second derivative.

c Complete the following tables of values:

x	0	1	2	3	4
$f'(x)$	3				

d Describe the rate of change of the gradient.

e Are there any points of inflection on $f(x)$? Explain your answer.

9 Consider the function $f(x) = -3x^2 + 12x + 2$.

a Find $f'(x)$.

b Find $f''(x)$.

c Describe the rate of change of the gradient.

d Are there any points of inflection on $f(x)$? Explain your answer.

10 For each of the following functions:

i State the values of x for which the graph of the function is concave up.

ii State the values of x for which the graph of the function is concave down.

iii Explain why there is no point of inflection.

a $y = \sqrt{x+1}$

b $y = -\sqrt{x-2}$

c $y = \dfrac{1}{x}$

d $y = \dfrac{1}{x^2}$

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11 Consider the function $y = (x+6)^3$.

a State the transformation that turns $y = x^3$ into $y = (x+6)^3$.

b Find the point of inflection of $y = x^3$.

c Find the point of inflection of $y = (x+6)^3$.

d Complete the following table of values:

x	-7	-6	-5
y'			
y''		0	

e Is the point of inflection, a horizontal point of inflection? Explain your answer.

f For what values of x is the graph concave up?

12 Consider the function $y = (x+5)^3$.

a State the transformation that turns $y = x^3$ into $y = (x+5)^3$.

b Find the point of inflection of $y = x^3$

c Find the point of inflection of $y = (x+5)^3$.

d Complete the following table of values:

x	-6	-5	-4
y'			
y''		0	

e Is the point of inflection, a horizontal point of inflection? Explain your answer.

f For what values of x is the graph concave up?

13 Consider the function $y = x^4 - 8x^3 - 9$.

a Find y''.

b Find the points of inflection.

c Complete the following table of values:

x	-2	0	2	4	6
y'					
y''		0		0	

d Classify each point of inflection as an ordinary or horizontal point of inflection.

e For what values of x is the graph concave up?

14 Consider the function $y = 4x^3 - 16x^2 + 4x + 6$.

a Find y''.

b Find the point of inflection.

c Is the point of inflection, a horizontal or an ordinary point of inflection?

d For what values of x is the graph concave down?

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15 Consider the function $y = 3x^3 + 6x^2 + 8x + 3$.

- **a** Find y''.
- **b** Find the point of inflection.
- **c** Is the point of inflection, an ordinary or horizontal point of inflection?
- **d** For what values of x is the graph concave down?

16 Consider the function $y = x^5 - 3x^2$.

- **a** Find y''.
- **b** Find the exact point of inflection.
- **c** Is the point of inflection an ordinary or a horizontal point of inflection?
- **d** For what values of x is the graph concave up?
- **e** For what values of x is the graph concave down?

17 The first derivative of a certain function is $f'(x) = 3x^2 + 9x$.

- **a** Determine the interval over which the function is increasing.
- **b** Determine the interval over which the function is decreasing.
- **c** Find $f''(x)$.
- **d** Determine the interval over which the function is concave up.
- **e** Determine the interval over which the function is concave down.
- **f** Find the x-coordinate of the maximum turning point.
- **g** Find the x-coordinate of the potential point of inflection.

Stationary points

18 Consider the function $f(x) = (x-5)^2 + 3$.

- **a** Find the x-coordinate of the turning point.
- **b** Find the value of the second derivative at this point.
- **c** Hence, classify the stationary point.

19 Consider the function $f(x) = (x-5)(x+4)^2$.

- **a** Find the turning points.
- **b** Find the second derivative.
- **c** Determine the nature of the turning points.
- **d** State the values of x where the graph of $f(x)$ is concave down

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20 Consider the function $y = x^4 + 4x^3 + 2$.

- **a** Find y'.
- **b** Find y''.
- **c** Find the stationary points.
- **d** Classify the stationary points.
- **e** Find the points of inflection.
- **f** Classify each point of inflection as ordinary or horizontal point of inflection.
- **g** For what values of x is the graph concave up?

21 Consider the function $f(x) = (x-8)(x-5)^2$.

- **a** Find $f'(x)$.
- **b** Find the turning points.
- **c** Find $f''(x)$.
- **d** Classify the turning points.
- **e** For what values of x is the graph concave down?

22 For each of the following functions:

- **i** Find the turning points.
- **ii** Classify the turning points.

- **a** $f(x) = x^3 - 27x - 7$
- **b** $f(x) = x^3 - 12x - 2$
- **c** $f(x) = 2x^3 + 9x^2 - 24x$
- **d** $f(x) = x^4 - 2x^2 + 3$

23 For each of the following functions:

- **i** Find the x-coordinates of the stationary points.
- **ii** Determine the nature of the stationary points.

- **a** $f(x) = 7x^3 + 8x^2 + 3x + 4$
- **b** $f(x) = 8(x-7)^2(x+2)$
- **c** $f(x) = -\dfrac{8}{x^2} + 2x$
- **d** $f(x) = 2x^3 - 30x^2 + 150x - 247$
- **e** $f(x) = 2(x-5)^3 + 1$

24 Consider the function $f(x) = 5x^3 + 7x^2 + 3x + 6$.

- **a** Find the x-coordinates of stationary points.
- **b** Find the x-value of the point of inflection.
- **c** How many stationary points are there?

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25 Consider the function $y = \left(x^2 - 5\right)^3$.

a Find y''.

b Find the points of inflection.

c Which of these points are horizontal points of inflection?

26 For each of the following functions:

i Find the stationary point.

ii Find the possible point of inflection.

iii Determine whether this point is a turning point or a point of inflection. Explain your answer.

a $y = 4 - (x - 5)^4$ **b** $y = 5 - (x - 4)^4$ **c** $y = 6 - (x - 8)^4$

27 The function $f(x) = ax^2 + \dfrac{b}{x^2}$ has turning points at $x = 1$ and $x = -1$.

a Use the fact that there is a turning point at $x = 1$ to form an equation for a in terms of b.

b Use the fact that there is a turning point at $x = -1$ to form an equation for a in terms of b.

c What can you deduce about the values of a and b?

28 The function $f(x) = ax^3 + bx^2 + 12x + 5$ has a horizontal point of inflection at $x = 1$.

a Use information about $f'(x)$ to write an equation involving a and b.

b Use information about $f''(x)$ to write an equation for b in terms of a.

c Hence, find the value of a.

d Hence, find b.

29 The function $f(x) = ax^3 + 18x^2 + cx + 4$ has turning points at $x = 4$ and $x = 2$. Find the value of:

a a **b** c

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